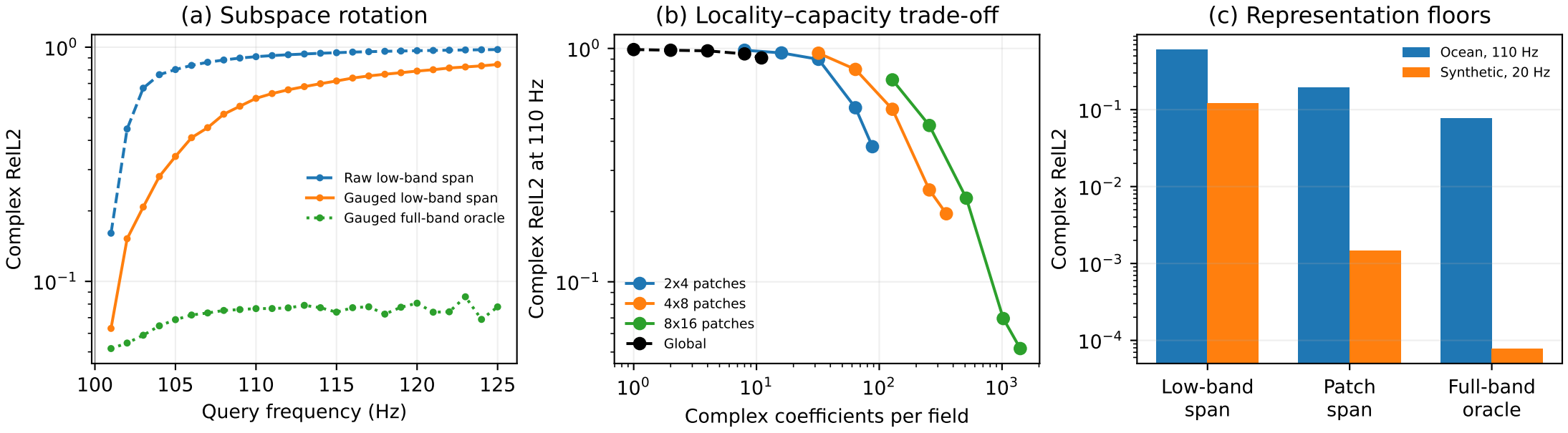
(a) Subspace rotation
Complex RelL2
Query frequency (Hz)
Raw low-band span
Gauged low-band span
Gauged full-band oracle
(b) Locality–capacity trade-off
Complex RelL2 at 110 Hz
Complex coefficients per field
2x4 patches
4x8 patches
8x16 patches
Global
(c) Representation floors
Complex RelL2
Ocean, 110 Hz
Synthetic, 20 Hz
Low-band span
Patch span
Full-band oracle